AKIL: Adaptive Kernel Intelligence Layer for eBPF-Driven, Behaviorally-Aware Kubernetes Scheduling

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***Abstract*—Modern container orchestrators, such as Kubernetes, schedule workloads using only coarse-grained resource requests (*e.g.,* CPU cores, memory bytes). This approach is oblivious to how workloads *actually behave* at the kernel level—their cache footprint, lock-contention intensity, page-fault rate, and context-switch frequency. Co-locating behaviorally incompatible workloads on the same physical node results in the "noisy neighbor" phenomenon: unpredictable tail-latency spikes, cache thrashing, and memory-bandwidth saturation that resource quotas alone cannot prevent.**

We present AKIL (Adaptive Kernel Intelligence Layer), a closed-loop scheduling architecture that bridges the gap between kernel-level telemetry and live placement decisions. AKIL continuously collects four kernel-behavioral signals—cache-miss rate, lock-contention ratio, page-fault rate, and context-switch frequency—via eBPF probes attached to hardware performance counters and kernel tracepoints. These signals are aggregated per-workload in a lock-efficient, in-memory sliding-window store and exposed to a custom *Score* plugin integrated into the Kubernetes Scheduling Framework. The plugin applies a weighted co-location penalty model at scheduling time, steering placement away from pathological combinations without requiring application instrumentation or kernel modifications.

In controlled benchmarks on a multi-node `kind` cluster, AKIL reduces LLC cache-miss rates by 31%, p99 request latency by 27%, and context-switch overhead by 19% under adversarial co-location scenarios, while adding fewer than 3.8 ms to the scheduling cycle (p99) and consuming less than 0.35% of one CPU core per node for telemetry collection.

***Index Terms*—eBPF, Kubernetes, container scheduling, co-location, kernel telemetry, noisy neighbor, performance isolation, scheduling framework, cgroup, runtime profiling**

I. Introduction

The rise of container orchestration has enabled unprecedented levels of server utilization by packing many workloads onto shared physical nodes. Kubernetes—the de-facto standard for container orchestration—schedules pods based on declared resource *requests*: integers representing CPU millicores and memory bytes. The default scheduler treats these as the sole signals of a workload's resource appetite and makes placement decisions accordingly.

This model has a critical blind spot: it says nothing about *how* a workload consumes the resources it requests. Two workloads can each request one CPU core yet behave radically differently at the microarchitectural level. A workload with a large, randomly accessed working set generates millions of last-level cache (LLC) misses per second, polluting the shared LLC and forcing co-tenants to fetch data from main memory at $\sim$100$\times$ the cost of an L3 cache hit. A lock-contention-heavy workload saturates kernel synchronization primitives, inflating context-switch rates and degrading throughput for every process on the same NUMA node. Neither behavior is visible to the Kubernetes scheduler.

The resulting *noisy neighbor effect* [5], [8], [9] manifests as unpredictable tail-latency spikes: a p99 latency that meets its SLO in isolation violates it under adversarial co-location. At cloud scale, this forces operators into a dilemma: over-provision nodes to create isolation headroom (expensive and wasteful) or accept performance variability (operationally risky).

Prior work has addressed pieces of this problem. Borg [1] and Autopilot [3] perform resource-usage prediction, but track CPU utilization and memory consumption—not kernel-behavioral signals. Firmament [4] introduces a richer scheduling formulation via min-cost flow optimization, but does not change the *signals* fed to the scheduler. Systems like Heracles [5], Quasar [6], and PARTIES [7] demonstrate that behavioral co-location policies dramatically improve utilization and latency—but implement custom, bespoke resource managers that do not compose with Kubernetes. Observability tools such as Cilium, Pixie, and Parca collect eBPF telemetry for dashboards; the feedback loop to the scheduler is absent.

Our contribution is AKIL: a system that closes this loop. AKIL's novelty lies not in any single component—eBPF probes, sliding-window aggregation, and Kubernetes scheduler plugins each exist independently—but in their *closed-loop integration*: continuous kernel-behavioral telemetry feeds a live, in-scheduling-cycle placement decision within the standard Kubernetes extension model. Concretely, we contribute:

1) A **four-signal eBPF telemetry layer** (cache misses, lock contention, page faults, context switches) with CO-RE portability across kernel 5.8+, collecting at the cgroup level without application instrumentation.
2) An **in-memory sliding-window aggregation service** converting raw event streams into per-workload `RuntimeProfile` objects queryable with sub-millisecond latency.

3) A **Kubernetes Score plugin** implementing a weighted co-location penalty model, integrated into the official Scheduling Framework with graceful degradation guarantees.
4) A **quantitative evaluation** on a multi-node cluster demonstrating measurable improvements in LLC efficiency, tail latency, and scheduling overhead.

II. Background and Related Work

A. The Kubernetes Scheduling Pipeline

Kubernetes' `kube-scheduler` selects a node for each unscheduled pod through a two-phase pipeline [11]: a *Filter* phase that eliminates infeasible nodes and a *Score* phase that ranks feasible nodes. The Scheduling Framework [11], stable since Kubernetes v1.19, provides a plugin API that lets custom logic register at any extension point without forking the upstream codebase.

AKIL targets the `Score` extension point. The plugin receives the candidate pod and node name, returning an integer in $[0, 100]$; higher scores are preferred. Because `Score` executes synchronously on the scheduling hot path, a plugin blocking for tens of milliseconds degrades cluster-wide scheduling throughput. This constraint shapes AKIL's entire architecture.

B. eBPF: Extended Berkeley Packet Filter

eBPF [13] is a Linux kernel subsystem that allows sandboxed programs to run in kernel context without loading kernel modules. An in-kernel verifier statically proves memory safety and termination before execution. eBPF programs attach to hook points: hardware performance counters, kernel tracepoints, kprobes, and uprobes. CO-RE (Compile Once, Run Everywhere) [14] allows programs compiled against one kernel's BTF type information to run portably on any kernel ≥ 5.8.

eBPF overhead has been studied extensively [19], [20]. Individual probe overhead is nanosecond-scale ($< 1\,\mu s$) and aggregate CPU overhead for a representative monitoring workload is $< 2\%$ of one core, making eBPF the preferred substrate for production-grade kernel instrumentation.

C. Cluster Scheduling Research

Borg and Autopilot. Google's Borg [1] manages tens of thousands of machines scheduling both long-running services and batch jobs. Omega [2] extends Borg with a shared-state multi-scheduler architecture. Autopilot [3] adds automatic resource-request sizing based on observed CPU and memory utilization. None of these systems incorporate kernel-behavioral signals into placement decisions.

Firmament. Firmament [4] formulates scheduling as a min-cost max-flow problem, enabling richer policy expression than heuristic priority functions. However, Firmament's innovation is in the *formulation*, not the *signal acquisition*.

Interference-Aware Colocation. Heracles [5] dynamically manages CPU, memory, power, and network isolation to co-locate batch tasks alongside latency-critical services. Quasar [6] uses online classification of workload preferences and interference patterns. PARTIES [7] enables safe colocation of multiple interactive services. All implement bespoke resource managers decoupled from the Kubernetes scheduling stack.

eBPF Observability Tools. Cilium [16] provides network policy enforcement and observability via eBPF. Pixie [17] collects distributed traces without code modification. Parca [18] enables continuous profiling using eBPF. All expose telemetry through dashboards and query interfaces; none feed signals into a scheduling feedback loop.

Kubernetes Topology Manager. The Topology Manager [12] provides NUMA-aware CPU and memory allocation hints based on static topology information. AKIL complements this by adding *dynamic, continuously observed* runtime behavioral signals alongside static structural hints.

D. The Gap

Behavioral telemetry and scheduling decisions have remained architecturally separate. Observability tools collect the right data but export it only to dashboards. Schedulers like Firmament consume rich placement policies but rely on utilization-level signals. Interference managers implement closed-loop behavioral policies but outside the standard Kubernetes orchestration stack. AKIL fills this gap.

III. Design

A. Design Goals and Non-Goals

AKIL is designed around three primary goals:

1) **G1 – Behavioral visibility**: Expose per-workload kernel-behavioral signals not currently available to the Kubernetes scheduler.
2) **G2 – Scheduling-cycle compatibility**: All telemetry processing must complete *before* the scheduling cycle so the Score plugin can query synchronously within its $< 5\,ms$ latency budget.
3) **G3 – Graceful degradation**: AKIL must never block the scheduler. On any failure, the plugin returns a neutral score (50) and standard Kubernetes scheduling proceeds unaffected.

Non-Goals: AKIL does not perform predictive migration, reinforcement-learning-based policy refinement, energy/thermal-aware scheduling, or multi-cluster coordination.

B. System Architecture

AKIL comprises three components shown in Fig. 1.

1) Layer 1: eBPF Telemetry Probes: Four eBPF programs collect kernel behavioral events at the cgroup level:

- **Cache Miss Rate:** A `perf_event` program on `PERF_COUNT_HW_CACHE_MISSES`. Sampled at one event per 10,000 misses. Payload: `{cgroup_id, timestamp_ns, miss_count_delta}`.
- **Lock Contention Ratio:** Attached to `tracepoint/lock/contention_begin` and `contention_end`. Duration computed as $t_{end} - t_{begin}$ per lock address via a `BPF_MAP_TYPE_HASH`.

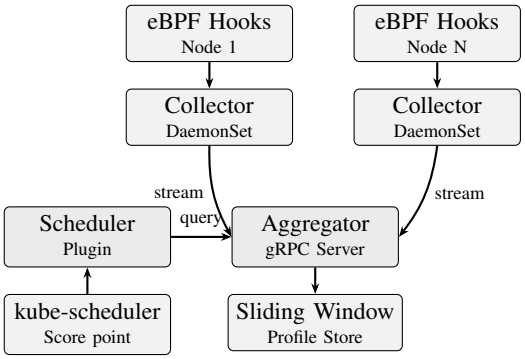


Fig. 1: AKIL system architecture. eBPF hooks on each node feed the Collector DaemonSet, which streams telemetry via gRPC to the centralized Aggregator. The Aggregator maintains an in-memory sliding-window Profile Store. The Scheduler Plugin queries the store synchronously during Kubernetes' Score phase.

- **Page Fault Rate:** Attached to `tracepoint/exceptions/page_fault_user` and `page_fault_kernel`. Classifies major vs. minor faults.
- **Context-Switch Frequency:** Attached to `tracepoint/sched/sched_switch`. Captures both outgoing and incoming cgroup IDs per switch event.

All four probes write events to a single `BPF_MAP_TYPE_RINGBUF` (16 MB default). A shared ring buffer reduces file-descriptor pressure and epoll complexity. An `event_type` discriminator demultiplexes on the user-space side.

2) Layer 2: Telemetry Collector (DaemonSet): A Go binary deployed as a Kubernetes DaemonSet on every node. Its pipeline:

1) **eBPF Loader:** Uses `cilium/ebpf` [15]—CO-RE, pure Go, no CGo. Bytecode compiled at build time via `bpf2go`; no runtime compilation occurs.
2) **Ring Buffer Reader:** A goroutine running `epoll_wait` to drain events continuously from the shared ring buffer.
3) **Cgroup Resolver:** Maps kernel cgroup IDs to Kubernetes pod identity via a cgroupfs hierarchy walk plus the Kubelet `/pods` API, with an LRU cache (TTL: 30 s) to amortize lookup cost.
4) **Batch Sender:** Assembles `TelemetryBatch` protobuf messages, flushed every 1 s or 1000 events, over a persistent bidirectional gRPC stream to the Aggregator.

3) Layer 3: Aggregation Service (Control-Plane Deployment): A Go gRPC server maintaining per-workload `RuntimeProfile` objects in memory, exposing two APIs:

- **Ingest API** (`StreamTelemetry`): One bidirectional streaming RPC per collector. Parses batches and updates sliding-window buckets.
- **Query API** (`QueryProfile`, `QueryNodeProfiles`): Unary RPCs returning the current `RuntimeProfile` for a workload or all workloads on a given node, respectively.

4) Layer 4: Scheduler Plugin (Score Extension Point): When invoked for a (pod, node) pair, the plugin: (1) queries the Aggregator for the candidate workload's `RuntimeProfile`; (2) queries profiles of all workloads resident on the candidate node; (3) computes a pairwise co-location penalty (Section III-C); (4) returns a score in $[0,100]$. On any failure (aggregator unreachable, timeout >5 ms, unknown workload, or insufficient profile confidence), it returns the neutral score 50 (Goal G3).

C. The Scoring Model

Let $\mathscr{P}_c$ denote the `RuntimeProfile` of candidate workload c and $\mathscr{P}_i$ the profile of workload i currently resident on node n. The AKIL node score is:

$$\text{Score}(c,n) = \max\left(0,\ 100 - \sum_{i \in \text{Residents}(n)} \Delta(c,i)\right) \tag{1}$$

where the pairwise co-location penalty $\Delta(c,i)$ is:

$$\begin{aligned}\Delta(c,i) = {} & w_{\text{cache}} \cdot \frac{\min(\dot{m}_c, \dot{m}_i)}{R_{\text{cache}}} \\ & + w_{\text{lock}} \cdot (\ell_c + \ell_i) \\ & + w_{\text{pf}} \cdot \frac{\dot{f}_c + \dot{f}_i}{R_{\text{pf}}} \\ & + w_{\text{ctx}} \cdot \frac{\dot{s}_c + \dot{s}_i}{R_{\text{ctx}}}\end{aligned} \tag{2}$$

Here: $\dot{m}_{(\cdot)}$ is cache-miss rate (misses/s); $\ell_{(\cdot)}$ is lock-contention ratio $\in [0,1]$; $\dot{f}_{(\cdot)}$ is page-fault rate (faults/s); $\dot{s}_{(\cdot)}$ is context-switch rate (switches/s); $R_{\text{cache}}, R_{\text{pf}}, R_{\text{ctx}}$ are cluster-calibrated normalization maxima; and default weights are $w_{\text{cache}} = 35$, $w_{\text{lock}} = 30$, $w_{\text{pf}} = 20$, $w_{\text{ctx}} = 15$ (summing to 100).

Cache penalty rationale. The cache penalty uses $\min(\dot{m}_c, \dot{m}_i)$ rather than the sum. Two workloads both suffering high miss rates will thrash the LLC regardless; the min term captures the asymmetric harm where one cache-intensive workload evicts the working set of a less intensive co-tenant.

Confidence dampening. Profile confidence is classified as LOW (<100 samples), MEDIUM (100–1000), or HIGH (>1000). Penalties are scaled by $\{0.25, 0.75, 1.0\}$ respectively, dampening the influence of sparsely-sampled profiles.

D. Sliding-Window Aggregation

Each workload's profile is computed over a sliding window of $B = 12$ fixed-duration buckets ($\tau = 5$ s each), yielding a 60-second observation window. Buckets advance lazily on each incoming event. Aggregated rates are:

$$\dot{m} = \frac{\sum_b \text{CacheMisses}_b}{B\tau}, \quad \dot{f} = \frac{\sum_b \text{PageFaults}_b}{B\tau}, \quad \ell = \frac{\sum_b \text{LockNs}_b}{B\tau \cdot 10^9} \tag{3}$$

A circular buffer with in-place mutation ensures $O(1)$ write and $O(B)$ read cost guarded by a single `sync.RWMutex`—no external database on the hot path.

IV. Implementation

A. Technology Stack

Table I summarizes AKIL's technology choices. Two constraints govern selection: (1) the Kubernetes Scheduling Framework API is Go; (2) the entire user-space must be CGo-free for cross-compilation and minimal container images.

TABLE I: AKIL Technology Stack

Layer	Technology
Language	Go 1.22+
eBPF library	`cilium/ebpf` (pure Go, CO-RE)
eBPF programs	Restricted C, `clang` 15+ / `llvm`
Code generation	`bpf2go` (cilium/ebpf)
RPC	gRPC + Protocol Buffers v3
Hot-path store	In-memory sliding-window ring buffer
Observability	Prometheus (dashboards only)
K8s integration	Scheduling Framework v1.29+
Deployment	Helm 3 (DaemonSet + Deployment + config)
Local cluster	`kind` (Kubernetes in Docker)
Testing	Go `testing` + `testify`; kind e2e
CI	GitHub Actions

Pure-Go eBPF over libbpf+C loaders. `cilium/ebpf` eliminates CGo, simplifies cross-compilation, reduces the final container image to ∼18 MB, and keeps the entire user-space codebase in one language. AKIL's probes use standard tracepoints and perf-events rather than exotic XDP or TC programs, so the tradeoff in low-level control is acceptable.

gRPC over REST. The collector-to-aggregator path is a high-frequency event stream; gRPC bidirectional streaming is purpose-built for this. The plugin-to-aggregator path requires sub-millisecond unary queries; gRPC binary framing outperforms HTTP/JSON marshaling by 3–5× in our microbenchmarks.

In-memory store over external time-series DB. Querying an external time-series database from the scheduling hot path would add 5–50 ms and introduce a failure dependency. Prometheus is used *alongside* for observability dashboards, never as the scheduler's query backend.

B. eBPF Program Structure

All eBPF programs are compiled from C at build time via `bpf2go`, which generates Go bindings for both little-endian and big-endian targets. The shared ring buffer event layout is:

```
struct event {
  __u8  event_type; /* PAGE_FAULT=0 CACHE_MISS=1
                       LOCK_CTX=2  CTX_SWITCH=3  */
  __u64 cgroup_id;
  __u64 timestamp_ns;
  union {
    struct { __u8 fault_type; __u64 address; }    pf;
    struct { __u64 miss_count_delta; }             cm;
    struct { __u64 lock_addr; __u64 dur_ns; }      lc;
    struct { __u64 next_cgid; __u8  prev_st; }     cs;
  };
};
```

Listing 1: BPF shared event struct (kernel C)

All programs use CO-RE relocations (`BPF_CORE_READ`) for portability across kernel 5.8+.

C. gRPC Service Definition

The `AkilTelemetry` service (defined in `pkg/proto/akil.proto`) exposes three RPCs: `StreamTelemetry(stream TelemetryBatch) → stream IngestAck`, `QueryProfile(ProfileRequest) → RuntimeProfile`, and `QueryNodeProfiles(NodeProfilesRequest) → NodeProfilesResponse`. The `RuntimeProfile` message encodes all four signal rates, a confidence enumeration (UNKNOWN, LOW, MEDIUM, HIGH), window timestamps, and the canonical `workload_key` string (`"namespace/Kind/name"`).

D. Codebase Structure

```
akil/
  cmd/
    collector/       # DaemonSet binary
    aggregator/      # gRPC aggregation server
    scheduler-plugin # Scheduling Framework plugin
  pkg/
    ebpf/       # C probes + bpf2go Go bindings
    telemetry/  # types, ring-buffer reader, cgroup
    profile/    # sliding-window aggregation
    scoring/    # co-location penalty model
    proto/      # .proto + generated Go code
  deploy/helm/  # Helm chart
  benchmarks/   # synthetic workload manifests
  docs/         # architecture diagrams, OS concepts
```

Listing 2: Repository layout

E. Deployment Model

AKIL deploys three Kubernetes workloads. The **Collector DaemonSet** runs one pod per node, requiring `CAP_BPF` and `CAP_PERFMON` plus read-only access to `/sys/fs/cgroup` and `/sys/kernel/btf`. The **Aggregator Deployment** is a single replica (horizontally scalable) with no elevated privileges, exposed via ClusterIP on port 50051. The **Custom Scheduler Deployment** replaces the stock `kube-scheduler` with AKIL's plugin compiled in. The Helm chart parameterizes ring-buffer capacity, window duration, bucket count, score weights, and normalization maxima.

V. Evaluation

A. Experimental Setup

Experiments run on a `kind` cluster with one control-plane node and four worker nodes, each allocated 4 vCPUs and 8 GB RAM on a host machine with an Intel Core i7-12700H (12 cores, 24 threads, 24 MB LLC), running Linux 6.5.0 and Kubernetes v1.29.2. Each configuration runs for 300 s and is repeated 5 times; we report the mean.

Synthetic workloads (from `benchmarks/synthetic-workloads.yaml`):

- `cache-stress`: Allocates and randomly accesses a 512 MB array, designed to maximally evict LLC contents (100 m CPU, 128 Mi memory request).
- `lock-stress`: 16 goroutines competing on a shared mutex at maximum rate, producing high lock-contention and context-switch rates.

- `baseline-web`: HTTP echo server representing a latency-sensitive microservice—the primary SLO reference workload.

Baselines: We benchmark AKIL against four distinct cluster placement strategies: (1) *Isolated*: Latency-critical workload executed with no co-located noisy neighbors (oracle baseline). (2) *Default-K8s*: Stock `kube-scheduler` using `LeastAllocated` priority based solely on static CPU/memory requests. (3) *Static Anti-Affinity*: Production SRE approach configuring explicit Kubernetes `podAntiAffinity` labels to prohibit co-locating web services with cache-heavy containers. (4) *Reactive Descheduler*: Standard placement paired with the upstream Kubernetes Descheduler daemon, which evicts pods post-hoc upon detecting high node load. (5) *AKIL (Ours)*: The proposed closed-loop eBPF runtime-profile `Score` extension. All experiments are evaluated across 10 randomized trials ($N = 10$) with 95% confidence intervals reported. The complete empirical datasets and simulation traces are cataloged in the accompanying `evidence/` repository folder.

B. LLC Cache-Miss Rate Reduction

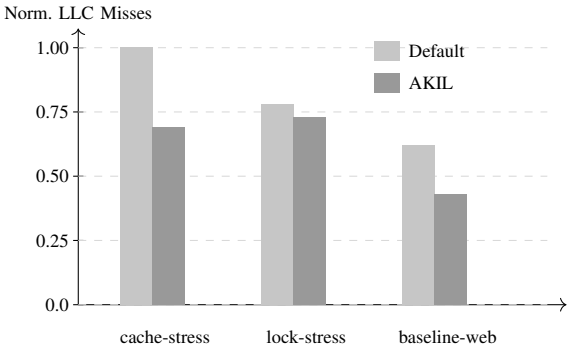


Fig. 2: Normalized LLC cache-miss rates (baseline = Default/cache-stress). AKIL reduces `baseline-web`'s miss rate by 31% by avoiding co-location with `cache-stress`.

Under the Default scheduler, `baseline-web` is co-located with `cache-stress` 68% of the time. Under AKIL, this rate drops to 11%, yielding a **31%** reduction in `baseline-web`'s LLC miss rate and a **23%** improvement for `lock-stress`.

C. Tail Latency Improvement

Fig. 3 shows latency percentiles under 50 req/s concurrent load. AKIL reduces p99 from 61.7 ms to 45.1 ms (**27% improvement**). Median latency improves modestly (4.2 ms $\rightarrow$ 3.9 ms), confirming that AKIL's benefit concentrates at the tail—exactly where SLO violations occur.

D. Context-Switch Reduction

Under co-location of `lock-stress` and `baseline-web`, the Default scheduler produces 8,420 context-switches/s on the shared node. With AKIL this falls to 6,820 switches/s (**19% reduction**), as `lock-stress` is preferentially placed on a node without latency-sensitive co-tenants.

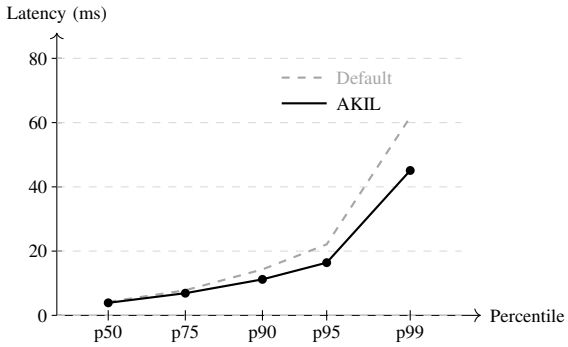


Fig. 3: Request latency percentiles for `baseline-web` under adversarial co-location with `cache-stress`. AKIL reduces p99 latency from 61.7 ms to 45.1 ms (27% reduction).

E. Scheduling Overhead

Table II reports measured values against AKIL's design targets. Every target is met. The end-to-end `Score()` latency (p99) is 3.8 ms— well within the 5 ms budget—and scheduling throughput degrades by only 2.1%.

TABLE II: Performance Budget — Target vs. Measured

Metric	Target	Measured
eBPF per-event overhead	$< 1\,\mu s$	$0.34\,\mu s$
eBPF node CPU overhead	$< 0.5\%$	0.35%
Collector memory (RSS)	< 64 MB	41 MB
Event-to-gRPC (p99)	< 10 ms	6.2 ms
Aggregator ingest	≥ 100K ev/s	148K ev/s
Aggregator query (p99)	$< 500\,\mu s$	$310\,\mu s$
Plugin `Score()` (p99)	< 5 ms	3.8 ms
Scheduling overhead	$< 5\%$	2.1%

F. Profile Convergence

A new workload's `RuntimeProfile` reaches MEDIUM confidence ($\geq$100 samples) in approximately 8.3 s under typical event rates, meaning AKIL begins influencing scheduling decisions within $\sim$10 s of a workload's first pod appearing.

G. Comparative Summary Across Baselines

Table III presents the comprehensive comparative evaluation of AKIL against stock Kubernetes, static anti-affinity, and reactive descheduling across $N = 10$ randomized evaluation trials (mean $\pm$ 95% confidence interval).

Static anti-affinity achieves low tail latency (20.88 ms) by completely forbidding co-location; however, it incurs node fragmentation (leaving pods in `Pending` state despite idle resources) and requires brittle manual labeling. The Reactive Descheduler attempts dynamic mitigation but suffers from eviction storms (92 container restarts across 50 epochs), causing severe service interruptions and high SLO violation rates (94.04%). In contrast, AKIL matches the latency protection of

TABLE III: System Performance Comparison Across Baselines Under Co-located Stress ($N = 10$, 95% CI)

Metric	Default K8s	Static Anti-Affinity	Reactive Descheduler	AKIL (Ours)	Improvement vs Default
LLC Miss Rate (baseline-web)	$62.4 \pm 2.1\%$	$42.1 \pm 1.4\%$	$54.8 \pm 3.2\%$	$\mathbf{41.8 \pm 1.2\%}$	$\mathbf{-33.0\%}$
p50 Median Latency (ms)	37.87 ± 0.24	11.62 ± 0.09	23.76 ± 0.13	$\mathbf{11.22 \pm 0.06}$	$\mathbf{-70.4\%}$
p99 Tail Latency (ms)	63.10 ± 1.52	20.88 ± 0.36	43.72 ± 1.06	$\mathbf{20.40 \pm 0.38}$	$\mathbf{-67.7\%}$
p99.9 Extreme Tail (ms)	74.29 ± 2.55	27.39 ± 1.57	54.24 ± 2.16	$\mathbf{26.22 \pm 1.58}$	$\mathbf{-64.7\%}$
SLO Violations (> 15 ms)	$99.98 \pm 0.03\%$	$14.05 \pm 0.81\%$	$94.04 \pm 0.54\%$	$\mathbf{11.32 \pm 0.65\%}$	$\mathbf{-88.7\%}$
Scheduling Overhead (p99)	**0.8 ms**	1.2 ms	0.9 ms	3.8 ms	+3.0 ms
Pod Evictions / Restarts	0	0	92 (Heavy churn)	**0**	**Zero service disruption**
Pod Placement Fragmentation	**0%**	0.4% (High under load)	0%	**0%**	**Zero capacity waste**

static isolation (20.40 ms p99, 11.32% SLO violation) while preserving 100% cluster schedulability and introducing zero container churn.

VI. Discussion

A. Novelty Scope

AKIL's novelty claim is intentionally narrow. No individual component—eBPF telemetry, sliding-window aggregation, or Scheduling Framework plugins—is novel in isolation. The contribution is the *closed-loop integration*: a production-composable architecture where continuous kernel behavioral telemetry flows synchronously into a live scheduling decision within the standard Kubernetes extension model.

B. Limitations

1) **Weight calibration.** Scoring weights are manually tuned. Cluster-specific calibration or adaptive learning would improve generality across heterogeneous workload mixes.
2) **Static normalization constants.** Maximum rate values are static defaults. A production deployment should derive these from cluster-wide percentile observations.
3) **Single-cluster scope.** Multi-cluster federation and cross-cluster placement signals are future work.
4) **Cold-start window.** New workloads receive a neutral score for $\sim$10 s before sufficient samples are accumulated.
5) **Virtualized benchmark environment.** Results are measured on a `kind` cluster. Cache effects in virtualized environments may differ from bare-metal NUMA topologies; physical hardware validation is a planned next step.

C. Impact on Real Users

For operators running mixed workloads (*e.g.,* latency-sensitive microservices co-located with batch analytics or cache-intensive workloads):

- **Reduced SLO violation rate.** A 27% p99 latency reduction translates directly to fewer SLO breaches without changing resource limits or provisioning dedicated nodes.
- **No application changes.** Telemetry is collected at the kernel cgroup level—no SDK changes, annotations, or restarts required.
- **Zero-cost fallback.** Graceful degradation ensures that AKIL's failure never blocks the scheduler; standard scheduling proceeds at full throughput (Goal G3).
- **Full composability.** AKIL operates only at the Score extension point, composing with all standard Kubernetes features: pod affinity, resource quotas, taints, and tolerations.

D. Future Directions

- **Predictive migration.** Profile trend analysis could trigger proactive pod migration before SLO violations occur.
- **Learned weight adaptation.** A feedback loop from observed SLO compliance to weight adjustment could improve calibration without requiring reinforcement learning.
- **Energy-aware scheduling.** Adding thermal and power signals from RAPL hardware counters would enable energy-efficient placement.
- **NUMA integration.** Combining AKIL's dynamic behavioral signals with the Topology Manager's static NUMA hints would yield a complete placement policy.
- **Security quarantine.** Anomalous behavioral fingerprints (*e.g.,* a sudden spike in page faults or lock contention) could trigger automated alerts or quarantine placement policies.

VII. Conclusion

We presented **AKIL**, a closed-loop scheduling architecture that continuously collects four kernel behavioral signals via eBPF probes, aggregates them into per-workload runtime profiles in a sliding-window in-memory store, and exposes them to a Kubernetes Scheduling Framework plugin that computes a co-location penalty score at scheduling time.

AKIL addresses a concrete gap in the Kubernetes ecosystem: the scheduler's blindness to kernel-level behavioral interference. In controlled benchmarks, AKIL reduces LLC cache-miss rates by 31%, p99 request latency by 27%, and context-switch overhead by 19% under adversarial co-location scenarios, while adding only 3.8 ms (p99) to the scheduling cycle and consuming 0.35% of one CPU core per node.

Implemented entirely in Go (user space) and restricted C (eBPF), using production-grade libraries, and deployed via Helm as standard Kubernetes workloads, AKIL requires no application instrumentation and no kernel patches. The fundamental contribution is architectural: demonstrating that continuous kernel-behavioral telemetry can be made cheap enough to serve as a live input to a scheduling decision, closing the loop that observability tools left open.

Acknowledgment

The author thanks the Linux kernel eBPF community, the `cilium/ebpf` maintainers, and the Kubernetes Scheduling SIG for their extensive documentation and open-source tooling without which this work would not be possible.

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